

# Oberseminar Data Science

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## Topic Presentation — Summer Semester 2026

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University of Passau · Chair of Data Science

# What is the Oberseminar?

The Oberseminar is the entry point for **Bachelor's and Master's theses** at the Chair of Data Science.

## Format

- Presentation and Q&A for students conducting a thesis at the chair
- Topic presentation: twice a semester
- In-between sessions: students presenting their topic, their final presentation and Q&A on the thesis.

## Who attends

- Students looking for a BSc or MSc thesis topic
- Students currently working on their thesis (progress updates)

## What today covers

- The registration process — step by step
- What we expect from you
- Defined topics (overview + link to full briefs)
- The open theme and how to apply

## No gos

- No thesis topics with companies
- No thesis topic outside of the suggested topics (except for excellent, research tier topics)
- No thesis topics that are not related to the research topics of the chair

Leave today with one topic in mind — and know exactly what to do next.

# Thesis Process

## Step 1 — Topic Selection

Pick one topic from the defined list, or prepare a one-pager for the open theme. Ask questions now. No guarantees you are getting a topic. We will select best on affinity to the chair (courses, grades).

## Step 2 — Entry Task

A 1–2 week scoped task assigned with the topic. It tests whether you and the topic are a good match.

## Step 3 — Research Exposé

A structured 4–6 page document (motivation, state of the art, research questions, methodology, work plan, limitations). Presented at the end-of-semester Oberseminar.

## Step 4 — Registration

Once the exposé is accepted, register with the examination office. **Registration is binding** — it starts the clock.



≡ You **cannot** register the thesis until the exposé has been accepted. A Master's thesis must be **submitted within six**

**months** of registration; Bachelor's theses have a shorter period per the examination regulations.

# What We Expect

## Scientific standard

- Clear research question — not a project description
- Related work that goes beyond tutorial summaries
- Reproducible experiments with baselines and metrics
- Honest limitations and scope boundaries

## Working style

- Self-directed — you drive the schedule
- Weekly or biweekly check-ins with your supervisor
- All code and data in a Git repository from day one
- Written communication in English

## The exposé — required structure

| Section            | Content                                   |
|--------------------|-------------------------------------------|
| Motivation         | Why this problem matters now              |
| State of the art   | What exists and where it falls short      |
| Research questions | At least 3 specific, answerable questions |
| Methodology        | Datasets, metrics, tools, evaluation plan |
| Work plan          | 3–5 milestones with realistic dates       |
| Limitations        | Known constraints and threats to validity |

**Limitations are mandatory.** A research plan without stated limitations signals insufficient understanding of the problem scope.

# Meeting Prof. Granitzer

## Weekly Q&A slot

- **Monday, 14:00–16:00** - presentation slots as defined in Stud.IP Abschlussarbeiten. You should be here to listen.
- Requires **prior registration through the chair's secretariat** — drop-ins are not possible
- Bring one concrete question or a short one-pager; unfocused check-ins get deferred
- For the open OUR<sup>2</sup>S / YOARS theme, align on scope in the Q&A **before** submitting the one-pager

## Contact

- Secretariat — to book the Q&A slot
- Email — [michael.granitzer@uni-passau.de](mailto:michael.granitzer@uni-passau.de)

## Where to find the full topic briefs

All defined topics live on the chair's master thesis-topics page. The slides that follow are pitch cards — the full briefs (research questions, methodology, why-it-fits) are maintained there and are the source of truth.

- Master page: </research/thesis-topics/>
- How to apply: </research/thesis-topics/overview/>

Read the master brief before the Q&A. The slot is for alignment and scope, not for retelling the brief.

# Defined Topics

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## Pitch Cards — Full Briefs on the Master Page

Each topic comes with a 1–2 week entry task. Completing it well is the condition for advancing to the exposé phase. Full research questions, methodology, and background live on the master thesis-topics page.

# Topic 1: Fast URL Risk Classifier

**Scope:** Bachelor's thesis or compact Master's project thesis

**Pitch:** A URL-only classifier that flags spam, ads, and security-risk URLs in microseconds — lightweight enough to run at web scale, data-driven enough to generalise beyond hand-written domain rules.

**Connects to:** OUR<sup>2</sup>S pipeline — fast URL-level filtering is a practical bottleneck before full document retrieval.

**Full brief:** </research/thesis-topics/#url-risk-classifier>

**Entry task:** Given a small labelled URL dataset, implement a character n-gram feature extractor and train a logistic regression classifier. Report precision, recall, and inference time per URL.

# Topic 2: Ranking Profiles with Client-Side XGBoost

**Scope:** Master's thesis or research prototype

**Pitch:** Expose a small family of named ranking profiles (e.g. "freshness-first", "authority-first", "diversity-first") that clients select at query time. Discover them without explicit preference labels. Deployable as a real OUR<sup>2</sup>S API feature.

**Connects to:** OUR<sup>2</sup>S search API — directly extends the chair's infrastructure.

**Full brief:** </research/thesis-topics/#ourrs-ranking-profiles>

**Entry task:** Using a provided OUR<sup>2</sup>S query log sample, cluster queries by result-feature distributions and identify two or three clearly distinct ranking biases. Describe each in plain language.



# Topic 3: Hierarchical Multilabel Classification for OpenWebIndex

**Scope:** Master's thesis

**Pitch:** Replace the current rule-based category assignment on the Open Web Index with a data-driven hierarchical multilabel classifier over the Curlie taxonomy. Compare Jina v5 embeddings against TF-IDF / BM25 baselines on both accuracy and throughput.

**Connects to:** Open Web Index — the dataset produced would be a public research contribution.

**Full brief:** </research/thesis-topics/#openwebindex-hierarchical-classification>

**Entry task:** Given 1000 labelled web pages from the Curlie dataset, train a TF-IDF logistic regression classifier and report precision@1 and precision@3 on a held-out set. Describe where it fails.

# Topic 4: Multi-Objective Alignment for LLM Fine-Tuning

**Scope:** Master's thesis

**Pitch:** Alignment is usually optimised as one scalar reward, but helpfulness, factuality, safety, and task success genuinely conflict. Map the Pareto front; evaluate whether post-hoc profile switching can serve multiple alignment regimes from one base model.

**Connects to:** Agentic AI — aligned agents must balance competing objectives in real deployments.

**Full brief:** </research/thesis-topics/#multi-objective-alignment>

**Entry task:** Using a small instruction-tuned model and two alignment metrics (e.g. helpfulness and factuality), run fine-tuning at three different objective weightings and report whether the trade-off is observable.

# Topic 5: REST API Design and Control Primitives for AI Agent Systems

**Scope:** Master's thesis

**Pitch:** Re-evaluate classic REST principles — HATEOAS, modularity, DRY, idempotency, least privilege, self-description — empirically for LLM agents that plan across calls. Extend with agent-control primitives (permission scoping, interruption hooks, state-transition constraints) that reduce unsafe or unrecoverable actions.

**Connects to:** OUR<sup>2</sup>S API and agentic search — benchmark spans both REST design and control-primitive angles.

**Full brief:** </research/thesis-topics/#rest-api-design-for-agents>

**Entry task:** Take a small REST API and add HATEOAS links to all responses. Run an LLM agent on two tasks: one it has seen before, one it has not. Compare completion rates and error types with and without HATEOAS. As a second step, add an explicit state machine and compare agent recovery behaviour.



# Topic 6: Minimal LLM Agents for API-Specific Tasks

**Scope:** Master's thesis

**Pitch:** General-purpose LLMs are expensive and brittle for API-bounded agent tasks. Study whether small fine-tuned models can reliably adhere to a specific API — and how minimal they can be while staying reliable on structured, API-specific workflows.

**Connects to:** Complements Topic 5 — same benchmark APIs, different angle: model optimisation vs. API design.

**Full brief:** </research/thesis-topics/#minimal-llm-agents-api>

**Entry task:** Fine-tune a small instruction-tuned model (1–3B parameters) on a synthetic dataset of correct API calls for one simple REST API. Compare completion rate and hallucinated-endpoint rate against a larger base model without fine-tuning.

# Open Theme

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## Apply with a One-Pager

Open themes have no fixed specification. You define the research direction within the theme; selection is based on the quality of your one-page proposal.

# Open Theme: Web Search with OUR<sup>2</sup>S / YOARS



## Context

The Chair of Data Science runs its own open-web search infrastructure built on the [Open Web Index](#). Unlike commercial search engines, the system is fully transparent: retrieval behaviour, ranking, and result provenance can be inspected, compared, and modified.

**OUR<sup>2</sup>S** (Open Web Index Retrieval Research System) and **YOARS** (Your Open Web Index Agentic Retrieval Search System) provide the infrastructure for experiments on real web-scale data.

## Why this is open

The research question — and therefore the thesis — is yours to define within the theme of RAG, agentic search, or web retrieval.

## Example directions

## Scope

Bachelor's thesis or Master's thesis

## How to apply

Submit a **one-page proposal** by email before the entry-task phase. The one-pager must include:

1. A concrete research question (not just a topic area)
2. Why you are the right person to work on it (background, skills, motivation)
3. A plausible evaluation strategy

**Selection is based on proposal quality.** A well-scoped question with a credible evaluation plan outweighs domain expertise alone.

## Infrastructure access

- Search API and data: [ourrs.eu/developers](https://ourrs.eu/developers)

# Next Steps

## Defined topics (Topics 1–6)

1. Pick one topic and read its full brief on the master page
2. Book the Q&A slot through the secretariat (Monday 14:00–16:00)
3. Express interest; receive the entry task
4. Complete the entry task (1–2 weeks)
5. If accepted: write the research exposé
6. Present the exposé at the end-of-semester Oberseminar
7. Register the thesis (starts the 6-month MSc clock)

**Master page:** </research/thesis-topics/>

**Contact:** [michael.granitzer@uni-passau.de](mailto:michael.granitzer@uni-passau.de)

## Open theme (OUR<sup>2</sup>S / YOARS)

1. Read the OUR<sup>2</sup>S infrastructure docs at [ourrs.eu](http://ourrs.eu)
2. Develop a concrete research question
3. Submit the one-pager (you have one try)
4. Selection is based on proposal quality and past performance at lectures at the chair.

One-pagers that describe a topic area without a concrete research question will not be considered.

One topic, with proper motivation, research gap, a core research question and expected outcome. Be concrete and concise.